

Inspectorio: Optimizing supply chain management with Vertex AI

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Inspectorio improves supply chain management processes for brands, retailers, and suppliers, enabling a more transparent and holistic view of operations.



About Inspectorio

Founded in 2016, Inspectorio is the leading AI-powered supply chain platform optimizing performance, building resilience, and providing intelligence across production chains for global brands, retailers, and their multi-tier suppliers.

Headquartered in the US, Inspectorio has offices in Vietnam, China, and Spain.

Industries: Technology

Location: Vietnam

**Google
Cloud
results**

- Improves latency by

Transferring
operational
data-logs

- 2.5x with
Vertex AI
- Saves costs by 60x with Vertex AI compared to other GenAI solutions available in the market
 - 95% accuracy rate in chatbot query response

for precise
optimization

Google Cloud (<https://cloud.google.com/>)

Vertex AI (<https://cloud.google.com/vertex-ai/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Document AI

(<https://cloud.google.com/document-ai>)

Vertex AI (<https://cloud.google.com/vertex-ai/>)

An efficient supply chain is the backbone of every successful retail business. This is why getting it right is so important. With more than 12,000 customers, including some of the largest brands and retailers around the world, Inspectorio (<https://www.inspectorio.com/>) aims to help businesses streamline their supply chain, leveraging data to deliver actionable insights. From quality management, compliance, and traceability, to tracking for on-time shipment, Inspectorio uses digital technology to transform supply chain operations.

"Our focus is in sourcing, ordering, and manufacturing. The key challenge in this part of the supply chain is usually the lack of visibility and collaboration between brands, retailers, suppliers and factories due to the low level of digitalization," explains Khanh Pham, Director of Data Science and Analytics at Inspectorio.



Inspectorio's supply chain management operations

Inspectorio launched in 2016 on a different cloud platform but as the company began to grow organically, it decided on a long-term solution that was not only reliable and scalable, but one that could deliver enterprise level security and compliance to meet its customers' requirements.

"As we delve into the future of supply chain management, Inspectorio is leading the industry by using AI to turn data into prescriptive and predictive solutions for our customers. Our AI-enabled platform will propel our partners toward operation excellence by improving how brands and retailers collaborate and improve the performance of their

supply chains," says Carlos Moncayo Castillo, CEO and Co-Founder at the company.

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*—Khanh Pham, Director
of Data Science and
Analytics, Inspectorio*

A data architecture built for the future of supply chain management

Inspectorio brings value to clients through four key steps. The first, is to visit clients' business processes to address any fragmented parts in their supply chain for a certain level of standardization. The second step is analysis, where Inspectorio provides visibility and insight into areas of opportunities. Further into the value ladder is automation, where it supports process automation by conducting a risk assessment, and builds a machine learning model to create an efficient automation process. And finally, it helps businesses optimize their supply chain with industry best practices.



Inspectorio employee inspecting a package

For analysis and automation, all of Inspectorio's applications are hosted on Google Cloud. It uses Cloud Storage (<https://cloud.google.com/storage>) to store its unstructured data, and BigQuery (<https://cloud.google.com/bigquery>) as its data lake.

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Pham says the company is leveraging data to focus on two new developments. The first is optimization and prediction through risk assessment, while the second is its virtual assistant model. Both these models are deployed on Vertex AI (<https://cloud.google.com/vertex-ai>).

"For optimization, Vertex AI is used for monitoring and identifying trends, and giving our clients better visibility into their business as a whole. It can also be easily scaled across multiple clients in a cost-efficient manner," explains Pham. The team also uses Vertex AI to support the development of its virtual assistant by letting it answer questions customers have through natural language querying, where it can then provide users with the data directly.

Inspectorio had experimented with various models prior to settling on Vertex AI, but faced a few roadblocks when moving from prototype to production. "With the different generative AI services, we found that latency was a problem, with long response times. It was also costly as we were

charged based on the number of queries made, making it impossible for us to scale the service to all customers."

With Vertex AI, Inspectorio has been able to see improvements across all areas of concern. Latency has improved by 2.5x, while the team has noticed a cost reduction of 60x compared to if they had gone with a different service provider. With the support of historical data, Inspectorio also found its chatbot accuracy to be at 95 percent. Because Inspectorio is set up on Google Cloud with enterprise level security and data protection policies, setting up Vertex AI was also a much more seamless process as it did not require additional security checks and compliance verification that would have been required with other GenAI solutions.

"We hope to leverage other built-in AI capabilities moving forward, such as Vision AI or Speech-to-Text on top of Vertex AI for our copilot feature in the future," says Pham.

Improving end-to-end supply chain transparency for easy traceability and sustainability audits

Inspectorio is already developing a new feature on its platform with [Document AI](https://cloud.google.com/document-ai) (<https://cloud.google.com/document-ai>) that will support a more sustainable future in retail. As consumers become more socially conscious, and government organizations start to impose transparency for traceability and sustainability, Inspectorio is offering

a new service that can help its customers trace every step of their supply chain, from the selection of material to the manufacturing process and delivery.

"We really appreciate all the support provided by the team at Google Cloud, for guiding us through the processes and helping us realize the power of technology through applying them in our various use cases and applications."

—Khanh Pham, Director
of Data Science and
Analytics, Inspectorio

A lot of documentation is required to fulfill this process, and Inspectorio is looking to leverage Document AI to read and extract information. "We are using Document AI to not only extract the information from these documents, but to validate and verify them. It will greatly reduce the manual workload that would otherwise be required," says Pham.

Despite having many projects underway, the team at Inspectorio is keeping busy by consistently experimenting for new solutions. "We really appreciate all the support provided by the team at Google Cloud, for guiding us through the processes and helping us realize the power of technology through applying them in our various use cases and applications," says Pham.